

Technical Specification: Archivematica & AtoM

Integration on WSL2/Docker

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This document describes the architecture and configuration required to integrate Archivematica (1.x) and AtoM (2.x) running on the same Docker/WSL2 host, with specific focus on enabling DIP uploads from Archivematica to AtoM including thumbnail generation.

For installation steps, see the [Runbook](#).

Quick Reference

Service URLs and credentials

Service	URL	Username	Password
Archivematica Dashboard	http://localhost:62080	test	test
Archivematica Storage Service	http://localhost:62081	test	test
AtoM	http://localhost:63001	demo@example.com	demo

These are the default dev/hack environment credentials. Change them before any production use.

SFTP access (archivists uploading transfers)

Archivists deposit transfer packages into the system over SFTP using any standard client (WinSCP, FileZilla, or `sftp` CLI).

Field	Value
Protocol	SFTP
Host	IP of the Windows host (or localhost if connecting from the same machine)
Port	6222
Username	archivista
Password	Archivista2026
Remote folder	/home/archivista/transfer-source-sigedi

Files placed in that remote folder appear immediately in the Archivematica transfer source.

1. Architecture Overview

Both systems run in isolated Docker containers and do not share a filesystem by default. The integration uses two **host bind mounts** that bridge the gap without requiring network transfers between containers.



The two shared directories

Directory	Host path	Purpose
transfer-source	C:\ArchivematicaDrop\transfer-source	Archivists drop transfer packages here; Archivematica reads them as a transfer source
transfer-share	C:\ArchivematicaDrop\transfer-share	Archivematica writes DIPs here; AtoM reads them from its own container mount

The installation scripts inject both paths into `docker-compose.override.yml` automatically. The host paths above assume a WSL2 host; on a native Linux server they differ — see the next section.

Environments: WSL2 vs native Linux server

The install scripts auto-detect the host and adapt. The table below summarizes what changes between a **WSL2 + Docker Desktop** workstation and a **native Linux server** (e.g. Ubuntu 22.04).

Aspect	WSL2 + Docker Desktop	Native Linux server
Host path base (TRANSFER_BASE_DIR)	/mnt/c/ArchivematicaDrop (Windows C: drive)	\$HOME/ArchivematicaDrop
transfer-source host path	/mnt/c/ArchivematicaDrop/transfer-source	\$HOME/ArchivematicaDrop/transfer-source
transfer-share host path	/mnt/c/ArchivematicaDrop/transfer-share	\$HOME/ArchivematicaDrop/transfer-share
Pipeline data volumes (am-pipeline-data, ss-location-data)	Often auto-created by Docker Desktop	Must exist first — make create-volumes / mkdir -p ~/.am/... (native dockerd does not auto-create bind-volume sources)

vm.max_map_count	Applied inside the docker-desktop distro via wsl.exe	Applied directly with sysctl + persisted in /etc/sysctl.conf
DIP rsync failure mode	exit 23 from DrvFS rejecting permission/timestamp preservation on /mnt/c	exit 23 from ownership/permission mismatch on the share dir

Two rules for a working cross-stack share, regardless of environment:

1. **Both installers must resolve to the same host transfer-share path.** They share the `TRANSFER_BASE_DIR` contract, so running both with defaults on the same machine is enough. On a native server, run **both as the same OS user** (the default base is `$HOME` -relative) or set `TRANSFER_BASE_DIR` explicitly on both:

```
TRANSFER_BASE_DIR=/srv/archivematica bash installation/install_archivematica.sh
TRANSFER_BASE_DIR=/srv/archivematica bash installation/install_atom.sh
```

2. **Fix the DIP rsync exit 23** by setting the Dashboard **Rsync command** (Section 2A) to skip attribute preservation, which resolves both the WSL DrvFS case and the native permissions case:

```
rsync --no-perms --no-owner --no-group --no-times --omit-dir-times -0 -r
```

2. Integration Configuration

Even with the infrastructure automated by the install scripts, the following settings must be applied manually in the web interfaces to complete the integration.

A. Archivematica Dashboard — AtoM DIP Upload

Navigate to **Administration** → **AtoM DIP upload** in the Archivematica Dashboard (<http://localhost:62080>).

Archivematica Dashboard - Ad...

Archivematica Dashboard -

VAS-5950-2024.pdf - Demo site

Recibidos (5) - marjingu@gmail

localhost:62080/administration/dips/atom/

Processing configuration

General

Failures

Storage locations

Processing storage usage

DIP upload

Atom

ArchivesSpace

PREMIS agent

REST API

Users

Handle server config

Language

Version

AtoM DIP upload

The settings below configure DIP uploading to AtoM.

Upload URL *

URL where the AtoM index.php frontend lives, SWORD services path will be appended.

Login email *

E-mail account used to log into AtoM.

Login password *

Password used to log into AtoM.

AtoM version *

Rsync target

The DIP can be sent with Rsync to a remote host before is deposited in AtoM. This is the destination value passed to Rsync (see man 1 rsync). For example: foobar.com:~/dips/.

Rsync command

If --rsync-target is used, you can use this argument to specify the remote shell manually. For example: ssh -p 22222 -l user.

Debug mode

Show additional details.

REST API key

Used in metadata-only DIP upload.

Save

Setting	Value	Notes
Upload URL	http://host.docker.internal:63001/index.php	Routes from the Archivematica container back through the host to AtoM's port. Using localhost here would resolve inside the Archivematica container and fail.
Login email	demo@example.com	AtoM administrator account
Login password	demo	AtoM administrator password
AtoM version	2.x	
Rsync target	/home/transfer-share-from-archivematica	Path <i>inside the Archivematica container</i> where the DIP is placed
Rsync command	rsync --no-perms --no-owner --no-group --no-times --omit-dir-times -0 -r	Prevents exit 23 failures from filesystem/permission mismatches (DrvFS on WSL, ownership on native Linux)
REST API key	(see below)	Required for authorization and metadata mapping

Getting the AtoM REST API key: In AtoM, go to **Admin** → **Users**, edit the administrator user, and copy the value from the **REST API key** field. If it is empty, click **Generate** to create one.

B. Archivematica Storage Service — Verify pipeline locations

Navigate to **Locations** in the Storage Service (<http://localhost:62081/locations/>) and confirm that the pipeline's transfer source and AIP storage locations are active and pointing to the expected paths.

Archivematica Dashboard - AdArchivematica Storage ServiceVAS-5950-2024.pdf - Demo siteRecibidos (5) - marjingu@gmail.comlocalhost:62081/locations/Archivematica Storage ServiceHomePipelinesSpacesLocationsPackagesAdministrationLog out

All Locations

Locations have a purpose, such as AIP storage or transfer source, and are where packages are stored. Each Location is associated with a pipeline and can only be accessed by that pipeline.

Show10entriesSearch

Purpose	Pipeline	Path	Description	Space	UUID	Usage	Enabled
AIP Recovery	Archivematica on d50a43b17886	/var/archivematica/storage_service/recover	Default AIP recovery	badfa217-0...	9ae502f2-0c2f-4764-8cef-db3c0dc61094	0B / unlimited	Enabled
AIP Storage	Archivematica on d50a43b17886	/var/archivematica/sharedDirectory/www/AIPsStore	Store AIP in standard Archivematica Directory	badfa217-0...	e9c100b2-fa52-4c62-bdc9cdbe876f896	188098B / unlimited	Enabled
Currently Processing	Archivematica on d50a43b17886	/var/archivematica/sharedDirectory	None	badfa217-0...	f66a4907-37b0-4fd2-a2ea-f05b5a58a3ee	0B / unlimited	Enabled
DIP Storage	Archivematica on d50a43b17886	/var/archivematica/sharedDirectory/www/DIPsStore	Store DIP in standard Archivematica Directory	badfa217-0...	e34070b3-9b53-439b-9c5a-b72fa2bcb7c2	936052B / unlimited	Enabled
Storage Service Internal Processing	No pipelines	/var/archivematica/storage_service	For storage service internal usage.	badfa217-0...	0112af7d-5398-48ed-a218-870873040584	0B / unlimited	Enabled
Transfer Backlog	Archivematica on d50a43b17886	/var/archivematica/sharedDirectory/www/AIPsStore/transferBacklog	Default transfer backlog	badfa217-0...	c88d8d2b-32aa-4d9d-9d05-e6785ef26a8f	0B / unlimited	Enabled
Transfer Source	Archivematica on d50a43b17886	/home	Default transfer source	badfa217-0...	c3b1adfc-9802-498e-a181-ec6479f854dd	0B / unlimited	Enabled

Showing 1 to 7 of 7 entries

C. AtoM Global Settings — SWORD deposit directory

Navigate to **Admin** → **Settings** → **Global** in AtoM (<http://localhost:63001>).

Archivematica Dashboard - Adm x Archivematica Storage Service x Global setting - Demo site x Recibidos (5) - marjingu@gmail.com x

localhost:63001/index.php/settings/global

Clipboard

- DIP upload
- Default page elements
- Default template
- Diacritics
- Digital object derivatives
- Finding Aid
- Global**
- Header customizations
- I18n languages
- Identifiers
- Inventory
- Markdown
- Permissions
- Privacy Notification
- Security
- Site information
- Storage service
- Treeview
- Uploads
- User interface labels
- Web analytics

Application version

2.10.1 - 197

The current version of the application

Check for updates

☐ No

☒ Yes

Enable automatic update notification

Search and browse

Presentation

Multi-repository

Permalinks

System

Integrations

Google Maps Javascript API key (for displaying dynamic maps)

SWORD deposit directory

/home/transfer-share-from-atom

Save

Setting	Value	Notes
SWORD deposit directory	/home/transfer-share-from-atom	Path <i>inside the AtoM container</i> where it looks for incoming DIPs

If the field is missing from the UI: The setting can be hardcoded in `apps/qubit/config/app.yml` as `sword_deposit_dir: /home/transfer-share-from-atom`. After editing, clear the Symfony cache (`php symfony cc`) and restart `atom_worker`.

3. End-to-End Workflow

Use this sequence to validate the integration from start to finish.

- Deposit a transfer** — Copy a folder containing PDFs or images to the transfer source:
 - Via SFTP: connect with the credentials above and upload to `/home/archivista/transfer-source-sigedi`
 - Directly on the host: copy to `C:\ArchivematicaDrop\transfer-source`
- Start a transfer in Archivematica** — Dashboard → **Transfer** tab → Standard transfer → Browse → select the folder → Start transfer.
- Process through the pipeline** — Approve each micro-service step until the **Upload DIP** micro-service appears. Select the configured AtoM upload target.

4. **Monitor logs** — If the upload fails or stalls, watch the relevant container logs side by side:

```
# Archivematica MCP Client (sends the DIP)
cd ~/src/archivematica/hack
docker compose logs -f archivematica-mcp-client

# AtoM Worker (receives DIP, generates thumbnails)
cd ~/src/atom
export COMPOSE_FILE="docker/docker-compose.dev.yml:docker/docker-
compose.override.yml"
docker compose logs -f atom_worker
```

5. **Verify in AtoM** — Navigate to <http://localhost:63001>. The descriptive record should be present and the digital object viewer should show the uploaded files with generated thumbnails.

4. Remote Setup: SSH/Rsync

The shared bind-mount architecture works only when both systems run on the same physical host. For distributed deployments (Archivematica and AtoM on separate servers), the DIP must travel over the network via Rsync/SSH.

Archivematica side (sender)

1. Generate an SSH key pair for the `archivematica-mcp-client` container user.
2. Set **Rsync target** to a remote SSH path: `user@atom-server.example.com:/path/to/deposit/`
3. If using a non-standard port or key: set **Rsync command** to `ssh -i /home/archivematica/.ssh/id_rsa -p 2222`

AtoM side (receiver)

1. Ensure `sshd` is running and reachable from the Archivematica server's IP.
2. Add the Archivematica public key to `~/.ssh/authorized_keys` of the receiving user.
3. Ensure the receiving user has write permission to the SWORD deposit directory. If the SSH user differs from the PHP-FPM user (`www-data`), use a shared group with `chmod g+s` and `775` permissions so the AtoM worker can still read the files.